

Dichen Li

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Research Interests

I study **cross-embodiment learning** and **reinforcement learning for robot locomotion** on humanoid, quadruped, and hexapod robots. My research aims to develop agents that can learn from collective robot locomotion trajectories, generalize across diverse physical embodiments, and scale effectively with data. I am also interested in developing versatile locomotion skills across multiple environments.

Education

University of California, San Diego <i>Master's Program in Electrical and Computer Engineering</i> GPA: 3.96/4.00 Research interests: cross-embodiment robot learning, embodiment adaptation	09/2024-03/2026 San Diego, CA
University of California, Berkeley <i>Exchange Program in Electrical Engineering and Computer Sciences</i> Research interests: reinforcement learning for robot locomotion	09/2023-06/2024 Berkeley, CA
Xi'an Jiaotong University (C9 & 985) <i>Bachelor of Engineering in Automation, Elite Program (top 5%)</i> GPA: 3.92/4.30 Ranking: 19/220 <i>Graduate from the highest department Qian Xuesen Honors College</i>	09/2020-06/2024 Xi'an, China

Selected Publications

- Bo Ai*, Liu Dai*, Nico Bohlinger*, **Dichen Li*** (co-first author), Tongzhou Mu, Zhanxin Wu, K. Fay, Henrik I. Christensen, Jan Peters, Hao Su.
Towards Embodiment Scaling Laws in Robot Locomotion.
Conference on Robot Learning (CoRL), 2025.
- Tomson Qu, **Dichen Li** (second author), Avidesh Zakhori, Wenhao Yu, Tingnan Zhang.
Versatile Locomotion Skills for Hexapod Robots.
International Conference on Intelligent Robots and Systems (IROS), 2024.

Awards

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| Honored Graduate Students, Xi'an Jiaotong University | 2024 |
| Highest Level Scholarship of Elite Program (top 0.5%), Xi'an Jiaotong University | 2023 |
| First Prize of China Undergraduate Mathematical Contest in Modeling, Shaanxi Province | 2022 |

Research Experiences

Embodiment Adaptation in Robot Locomotion Project <i>Graduate Research Assistant advised by Prof. Hao Su</i>	04/2025-Present SU Lab, UC San Diego
Trained a locomotion policy for robots with continuously varying embodiment parameters and designed an adaptation module to infer unknown parameters and enhance robustness.	
Large-scale Embodiment Scaling Law Project <i>Graduate Research Assistant advised by Prof. Hao Su</i>	10/2024-05/2025 SU Lab, UC San Diego
- Developed a framework to distill locomotion policies from 1,000 diverse robot embodiments into a single generalizable policy, evaluated in Isaac Lab and deployed in real-world on Go2/H1. Discovered that training on accumulated embodiment improves performance, and that embodiment scaling outperforms data scaling. - Discovered that training on accumulated embodiment improves performance, and that embodiment scaling outperforms data scaling.	
Hexapod Robot Locomotion Skills Project <i>Undergraduate Research Intern advised by Prof. Avidesh Zakhori</i>	09/2023-05/2024 UC Berkeley
Trained and distilled 3 reinforcement learning policies for hexapod locomotion, and deployed them on real robots, achieving stair climbing, cave-like passages traversal (height from 22cm to 10cm), and obstacle avoidance (40cm blocks).	
Four-rotor UAV Control and Navigation Project <i>Undergraduate Research Assistant</i>	04/2022-08/2023 Xi'an Jiaotong University
Developed real-time UAV flight control with multi-level PID in C/Keil; integrated indoor localization via ROS and hector_mapping; and implemented object detection in Python using YOLOv5.	

Skills

- Coding & Tools: Python, PyTorch, C, MATLAB, Java, Linux, Bash, LaTeX, Git, Docker, Huggingface
- Robotics & robots: ROS, ROS2, Isaac Lab, Mujoco, SLAM, Unitree GO2, Unitree H1, Raspberry Pi
- Language: English (fluent), Chinese (native)